

The Builder's Workshop: A Non-technical Guide to Building Your First AI Agent

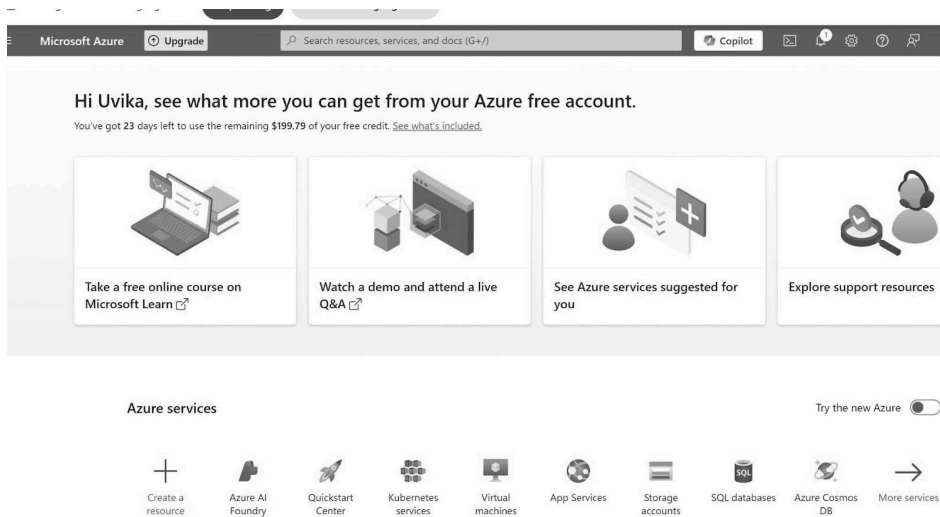
AGENTIC | VOICE & AI
Arlington, VA

October 27th, 2025

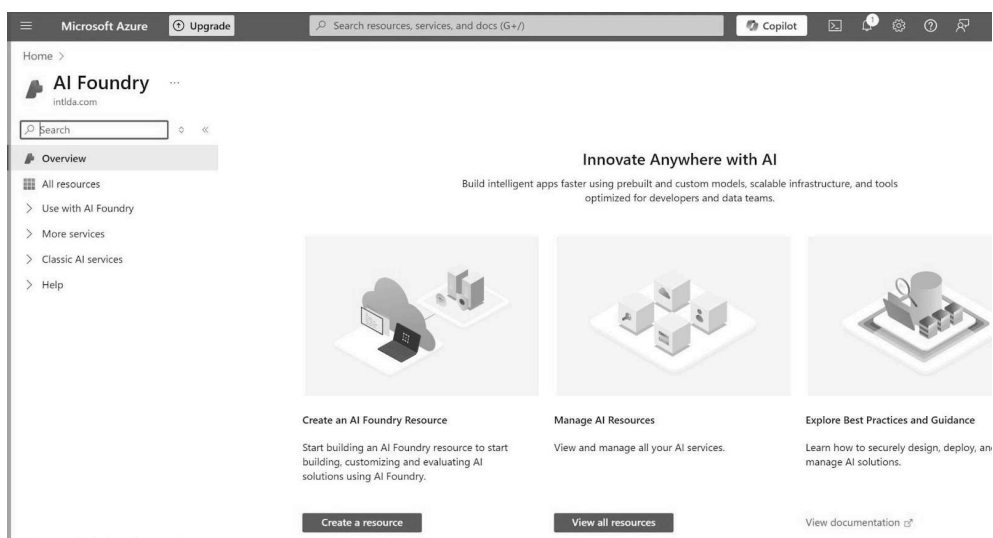
Padmini Soni, Rezonance AI
Uvika Sharma, INTLDA

Guide to building your first Agent in Azure AI Foundry

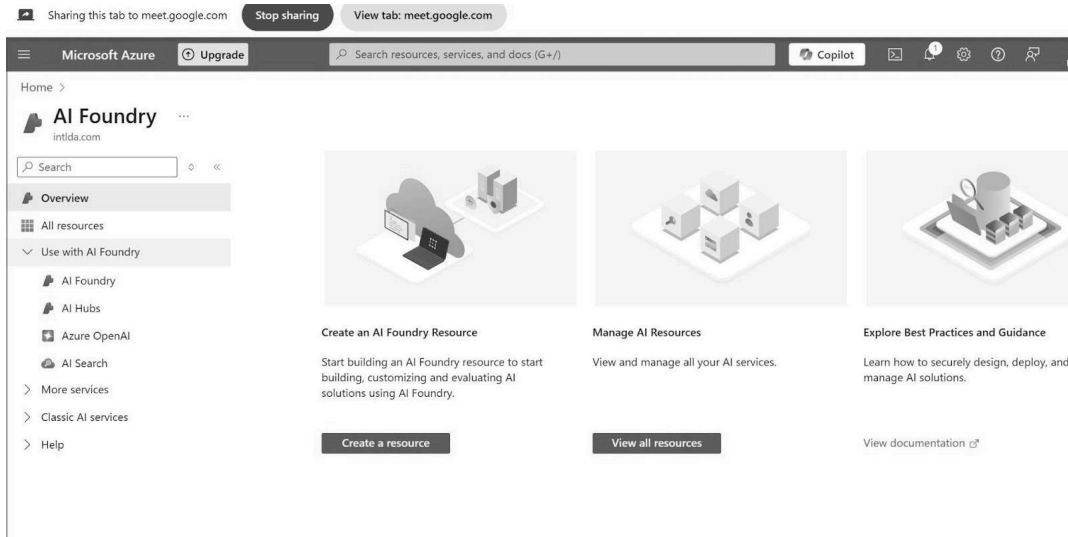
1. Go to <https://portal.azure.com/>



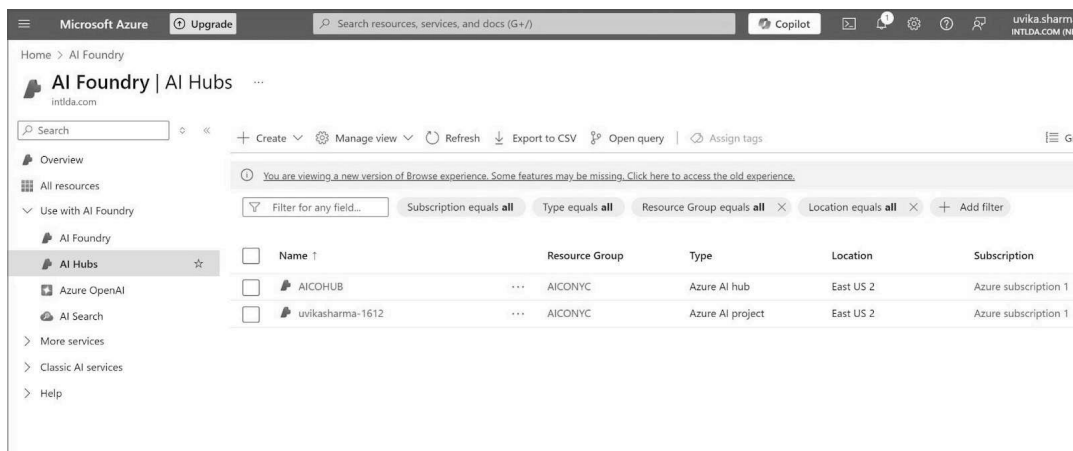
2. Click on Azure AI Foundry and you should see a screen below:



3. In the left navigation click on “Use with AI Foundry”



4. Click on “AI Hubs”



5. Create a Hub, by clicking on “Create” in the top navigation & selecting “Hub”

Home > AI Foundry

AI Foundry | AI Hubs

intida.com

Search

Overview

All resources

Use with AI Foundry

- AI Foundry
- AI Hubs**
- Azure OpenAI
- AI Search
- More services
- Classic AI services
- Help

+ Create Manage view Refresh Export to CSV Open query Assign

Project
Collaborate, organize, and track work to build AI apps.

Hub
Grouping container for projects. Provides security, connectivity, and compute management.

features may be missing. Click here to access the o

Type equals all Resource Group equ

	Resource Group	Type
☐	AICOHUB	Azure AI f
☐	uvikasharma-1612	Azure AI p

6. Create a new Resource Group. You can type in “<yourname>-rg”

Home > AI Foundry | AI Hubs >

Azure AI hub

Create an Azure AI hub resource

Basics Storage Inbound Access Outbound Access Encryption Identity Tags Review + create

Organization

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources. An AI hub is a collaboration environment for a team to share project work, model endpoints, compute, (data) connections, security settings, govern usage.

Subscription * Azure subscription 1

Resource group * [Create new](#)

Region *

Resource details

Name *

Friendly name

Default project resource group

Same as hub resource group ☐

A resource group is a container that holds related resources for an Azure solution.

Name *

OK Cancel

7. Add the Hub name. You can type in “<yourname>-hub”. You can add in a Friendly anime for your hub. Do not change any default values.

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > AI Foundry | AI Hubs >

Azure AI hub

Create an Azure AI hub resource

Basics Storage Inbound Access Outbound Access Encryption Identity Tags Review + create

Organization

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources. An AI hub is a collaboration environment for a team to share project work, model endpoints, compute, (data) connections, security settings, govern usage.

Subscription * Azure subscription 1

Resource group * (New) Uvika-rg
Create new

Region * East US 2

Resource details

Name * uvika-hub

Friendly name Uvika hub

Default project resource group

Same as hub resource group ☒

8. Click on “Next:Storage”. Leave all the default values as they are.
9. Click on “Next: Inbound Access”. Leave all the default values as they are.
10. Click on “Next: Outbound Access”. Leave all the default values as they are.
11. Click on “Next: Encryption”. Leave all the default values as they are.
12. Click on “Next: Identity”. Leave all the default values as they are.
13. Click on “Next: Tags”. We are not creating any tags for now.
14. Click on “Next: Review + Create”.
15. Click on “Create”

Home > AI Foundry | AI Hubs >

Azure AI hub

Create an Azure AI hub resource

Validation passed

Basics Storage Inbound Access Outbound Access Encryption Identity Tags Review + create

Basics

Subscription	Azure subscription 1
Resource group	(New) Uvika-rg
Name	uvika-hub
Default project resource group	(New) Uvika-rg

Resources

Region	East US 2
AI Services	(new) uvikahub1302783727
Storage account	aicohub8743675602 (AICONYC)
Key vault	(new) uvikahub4166505363
Application insights	None
Container registry	None

Create < Previous Next > Download a template for automation

16. Your AI Hub will be deployed shortly and you will get a message saying “Your deployment is complete”

The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with 'Microsoft Azure', an 'Upgrade' button, a search bar, and a 'Copilot' button. Below this, the breadcrumb 'Home >' is followed by 'Microsoft.MachineLearningServices | Overview'. The main content area is titled 'Deployment' and shows a progress indicator '... Deployment is in progress'. It lists deployment details: 'Deployment name : Microsoft.MachineLearningServices', 'Subscription : Azure subscription 1', 'Resource group : Uvika-rg', 'Start time : 6/2/2025, 3:47:59 PM', and 'Correlation ID : 36c519ba-5b08-4eb3-b617-0931e7ba3d71'. A table titled 'Deployment details' shows three resources: 'uvika-hub' (Microsoft.MachineLearningServ), 'uvikahub1302783727' (Microsoft.CognitiveServices/acc), and 'uvikahub4166505363' (Key vault). The right sidebar contains links to 'Microsoft Defender', 'Free Microsoft', and 'Work with an expert'.

Resource	Type	Status	Operation details
uvika-hub	Microsoft.MachineLearningServ	Accepted	Operation details
uvikahub1302783727	Microsoft.CognitiveServices/acc	Created	Operation details
uvikahub4166505363	Key vault	OK	Operation details

17. Click on “Go to resource”

The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with 'Microsoft Azure', an 'Upgrade' button, a search bar, and a 'Copilot' button. Below this, the breadcrumb 'Home >' is followed by 'Microsoft.MachineLearningServices | Overview'. The main content area is titled 'Deployment' and shows a progress indicator '✓ Your deployment is complete'. It lists deployment details: 'Deployment name : Microsoft.MachineLearningServices', 'Subscription : Azure subscription 1', 'Resource group : Uvika-rg', 'Start time : 6/2/2025, 3:48:24 PM', and 'Correlation ID : 36c519ba-5b08-4eb3-b617-0931e7ba3d71'. A table titled 'Deployment details' shows three resources: 'uvika-hub' (Microsoft.MachineLearningServ), 'uvikahub1302783727' (Microsoft.CognitiveServices/acc), and 'uvikahub4166505363' (Key vault). The right sidebar contains links to 'Cost management', 'Microsoft Defender for Cloud', 'Free Microsoft tutorials', and 'Work with an expert'.

Resource	Type	Status	Operation details
uvika-hub	Microsoft.MachineLearningServ	Accepted	Operation details
uvikahub1302783727	Microsoft.CognitiveServices/acc	Created	Operation details
uvikahub4166505363	Key vault	OK	Operation details

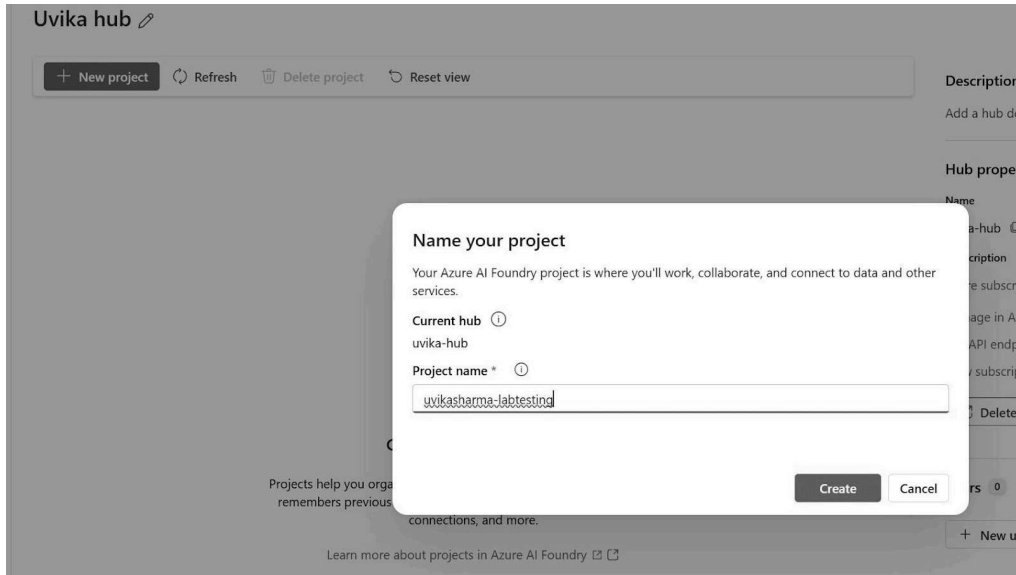
18. Click on “Launch Azure AI Foundry”

The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with 'Microsoft Azure', an 'Upgrade' button, a search bar, and a 'Copilot' button. Below the navigation bar, the breadcrumb trail reads 'Home > Microsoft.MachineLearningServices | Overview >'. The main header area displays 'uvika-hub' with a star icon and 'Azure AI hub' below it. A search bar and three action buttons ('Create project', 'Download config.json', 'Delete') are visible. On the left, a sidebar lists navigation options: Overview (selected), Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Events, Settings, Monitoring, Automation, and Support + troubleshooting. The main content area is divided into two columns. The left column, titled 'Essentials', lists key attributes: Resource group (uvika-rg), Location (East US 2), Subscription (Azure subscription 1), Subscription ID (51c939a7-f1e7-4f95-9ec7-46b4e097c41c), and Key Vault (uvikahub4166505363). The right column lists additional attributes: Project resource group (uvika-rg), Storage (aicohub8743675602), Container Registry (edit), Application Insights (edit), and Provisioning State (Succeeded). Below this, a section titled 'Govern the environment for your team in AI Foundry' features a large 'A' icon and text explaining the benefits of the Azure AI Foundry, with a 'Launch Azure AI Foundry' button.

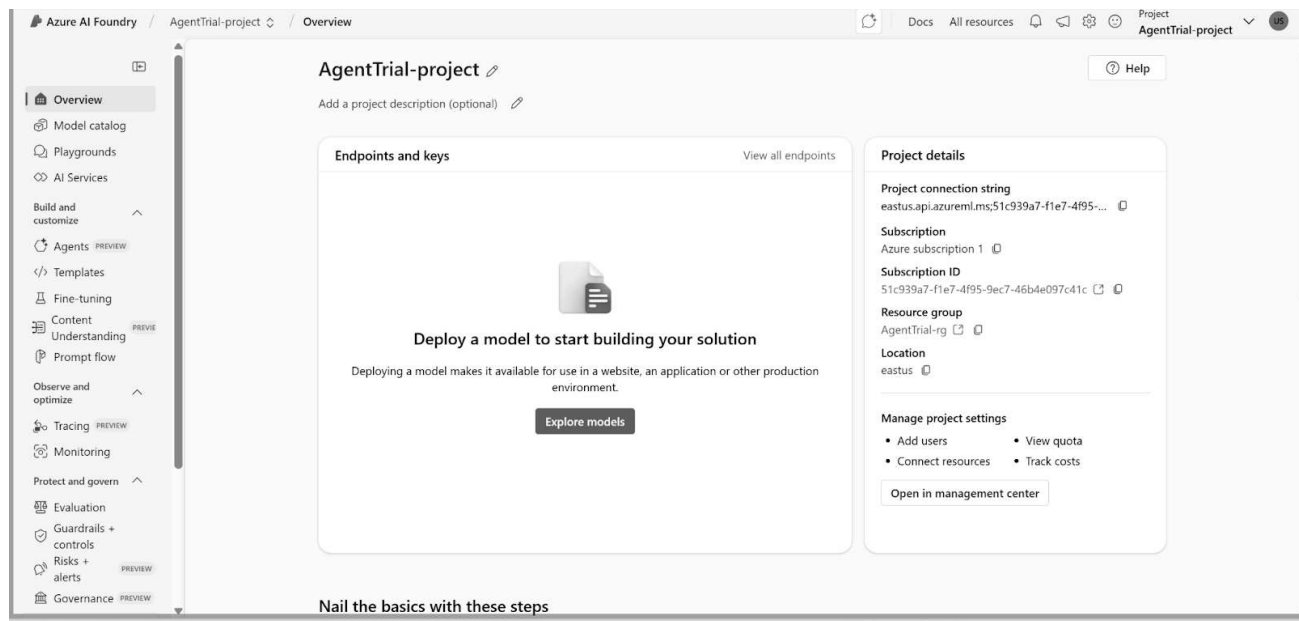
20. We are now in our AI Hub space. Click on “New Project”

The screenshot shows the 'Uvika hub' management center within the Azure AI Foundry. The breadcrumb trail at the top reads 'Azure AI Foundry | Management center / Hub overview'. On the left, a sidebar for the 'Management center' lists options: All resources, Quota, Hub (uvika-hub), Overview (selected), Users, Models + endpoints, Connected resources, and Compute. The main content area has a header 'Uvika hub' with an edit icon. Below the header is a toolbar with four buttons: 'New project' (highlighted), 'Refresh', 'Delete project', and 'Reset view'. The central area features a large document icon and the text 'Create your first project', followed by a paragraph explaining that projects help organize AI assets and save state, and a link to 'Learn more about projects in Azure AI Foundry'. On the right, there's a 'Description' section with a text area, a 'Hub properties' section listing details like Name (uvika-hub), Location (eastus2), Subscription (Azure subscription 1), and Resource Group (Uvika-rg), and a 'Users' section with a 'New user' button.

21. A pop-up will open. Give a project name. You can type in “<yourname>-project” and click on “Create”

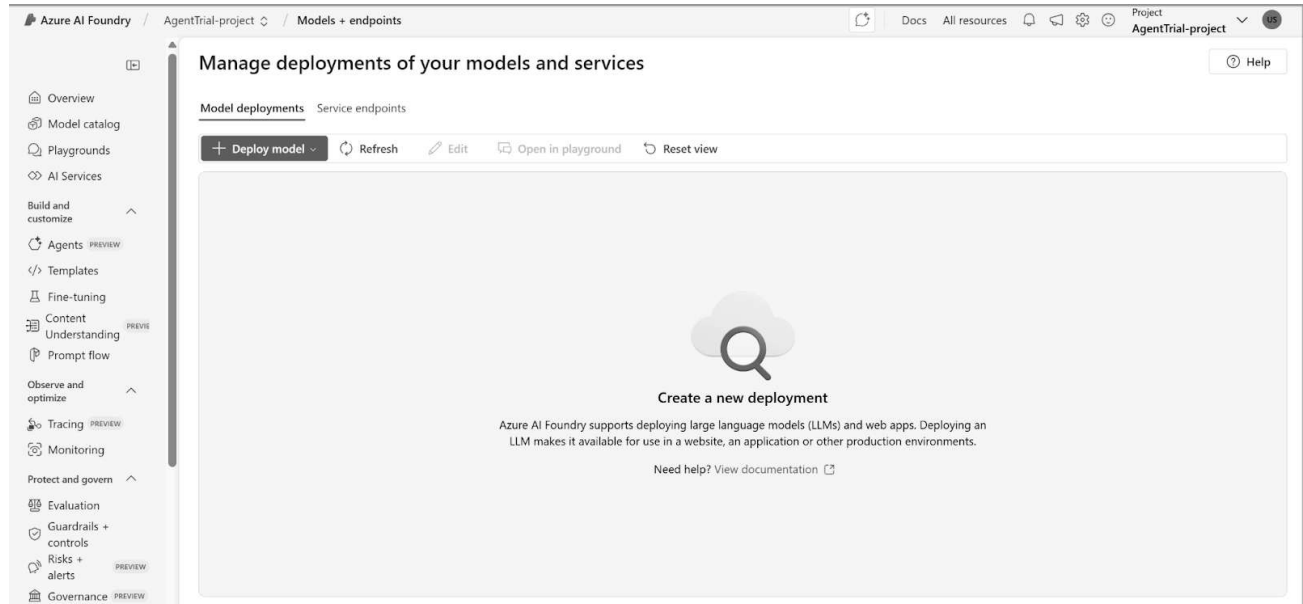


22. You will now be presented with a screen with endpoints, keys and project details



23. View all end points

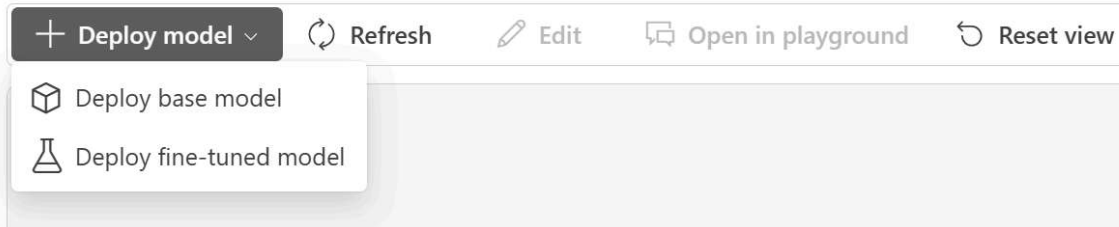
24. Click on Deploy a Model



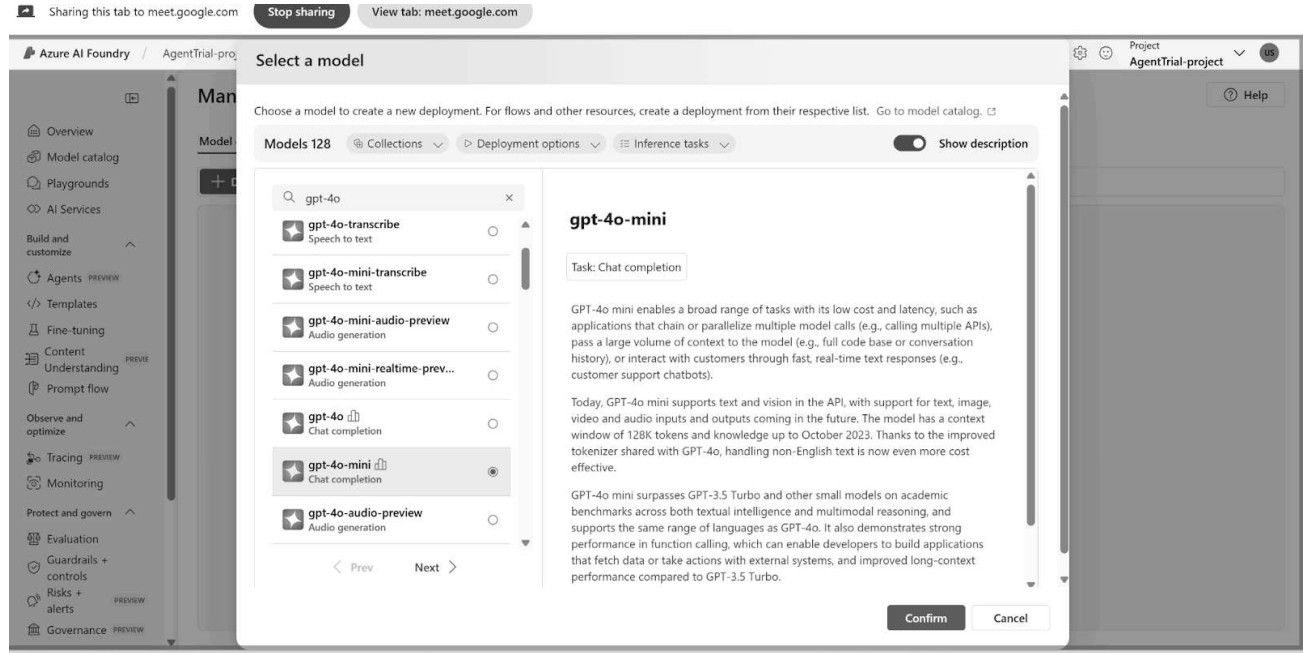
25. Select Deploy a Base Model

Manage deployments of your models and services

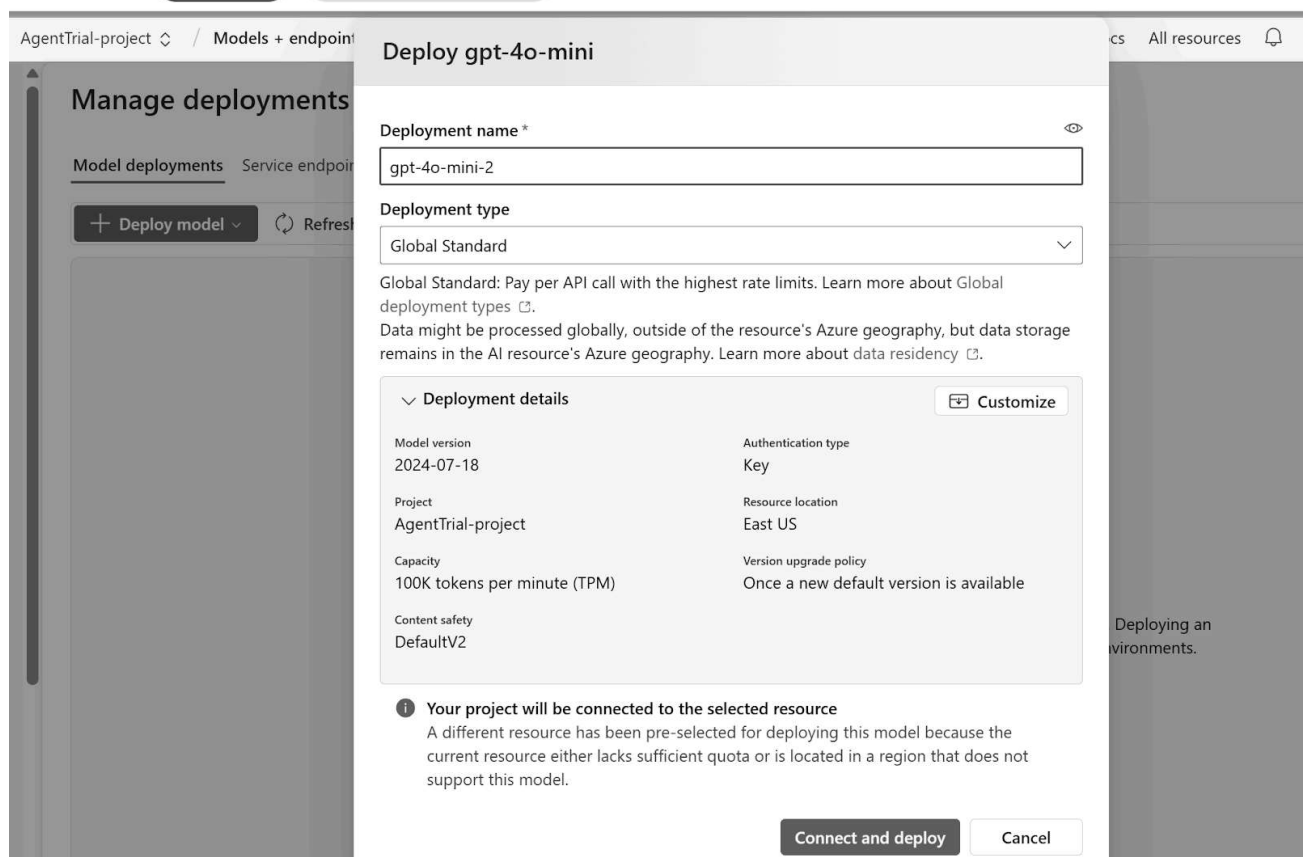
Model deployments Service endpoints



26. Select gpt-4o-mini



27. Select Deployment name



28. You will now get a screen like this:

Sharing this tab to meet.google.com Stop sharing View tab: meet.google.com

Azure AI Foundry / AgentTrial-project / Models + endpoints / gpt-4o-mini-2

Project: AgentTrial-project

gpt-4o-mini-2

Details Metrics

Open in playground Request quota Edit Delete

Endpoint

Target URI
https://aicotest-hub.openai.azure.com/openai/deployments/...

Authentication type
Key

Key
.....

Deployment info

Name	gpt-4o-mini-2	Provisioning state	Succeeded
Deployment type	Global Standard	Created on	2025-10-22T21:09:55.3500455Z
Created by	635bde24-66c8-4816-b855-078c0184aa17	Modified on	Oct 22, 2025 5:09 PM
Modified by	635bde24-66c8-4816-b855-...	Version upgrade policy	Once a new default version is

Language: Python SDK: Azure OpenAI SDK Authentication type: Key Authentication

Open in VS Code

Get Started

Below are example code snippets for a few use cases. For additional information about Azure OpenAI SDK, see full documentation and samples.

1. Authentication using API Key

For OpenAI API Endpoints, deploy the Model to generate the endpoint URL and an API key to authenticate against the service. In this sample endpoint and key are strings holding the endpoint URL and the API Key.

The API endpoint URL and API key can be found on the Deployments + Endpoint page once the model is deployed.

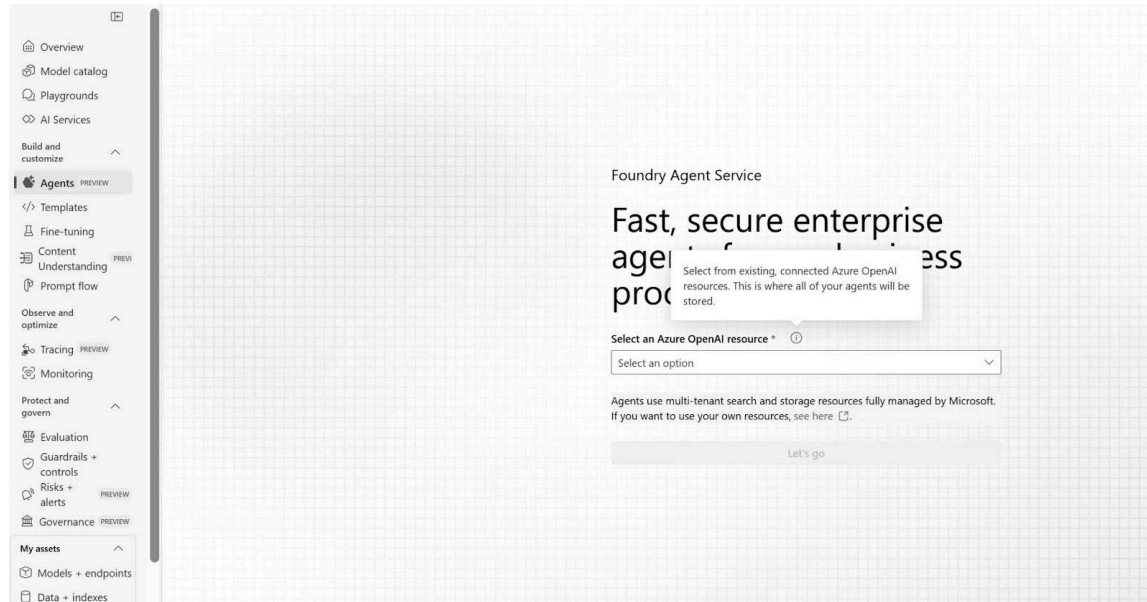
To create a client with the OpenAI SDK using an API key, initialize the client by passing your API key to the SDK's configuration. This allows you to authenticate and interact with OpenAI's services seamlessly:

```
import os
from openai import AzureOpenAI

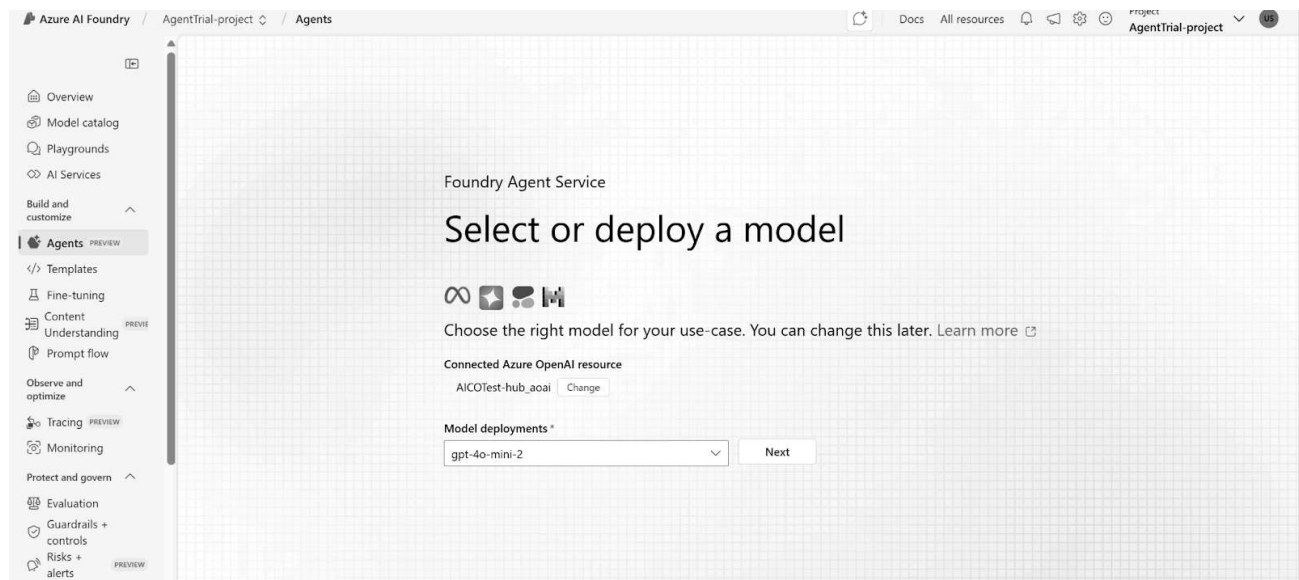
client = AzureOpenAI({
```

29. We are now ready to create our first agent! Click on “Agents” in the left navigation

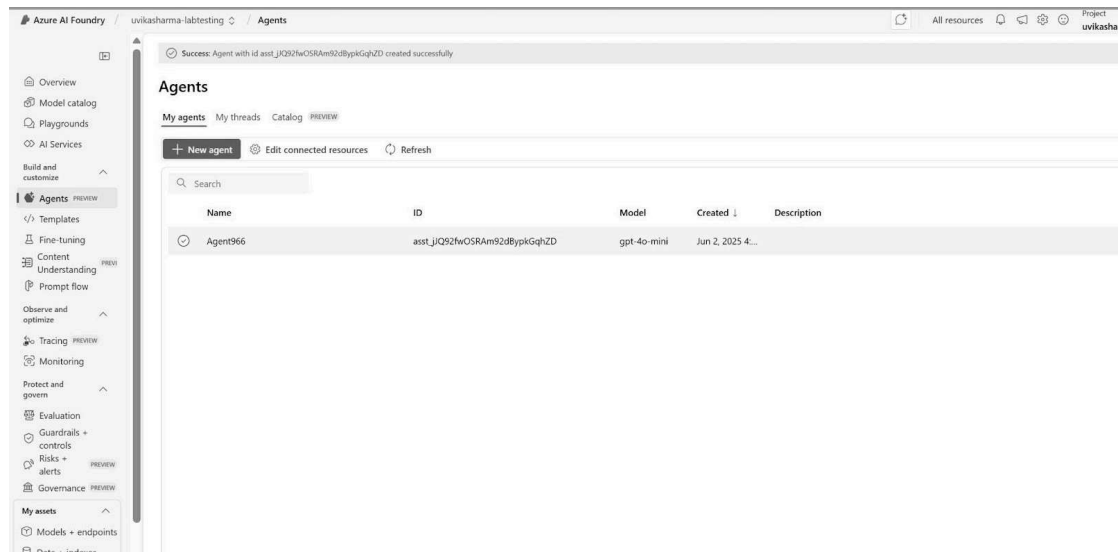
30. Select the AOAI resource. This is where all your agents will be stored. Click on “Let’s go”



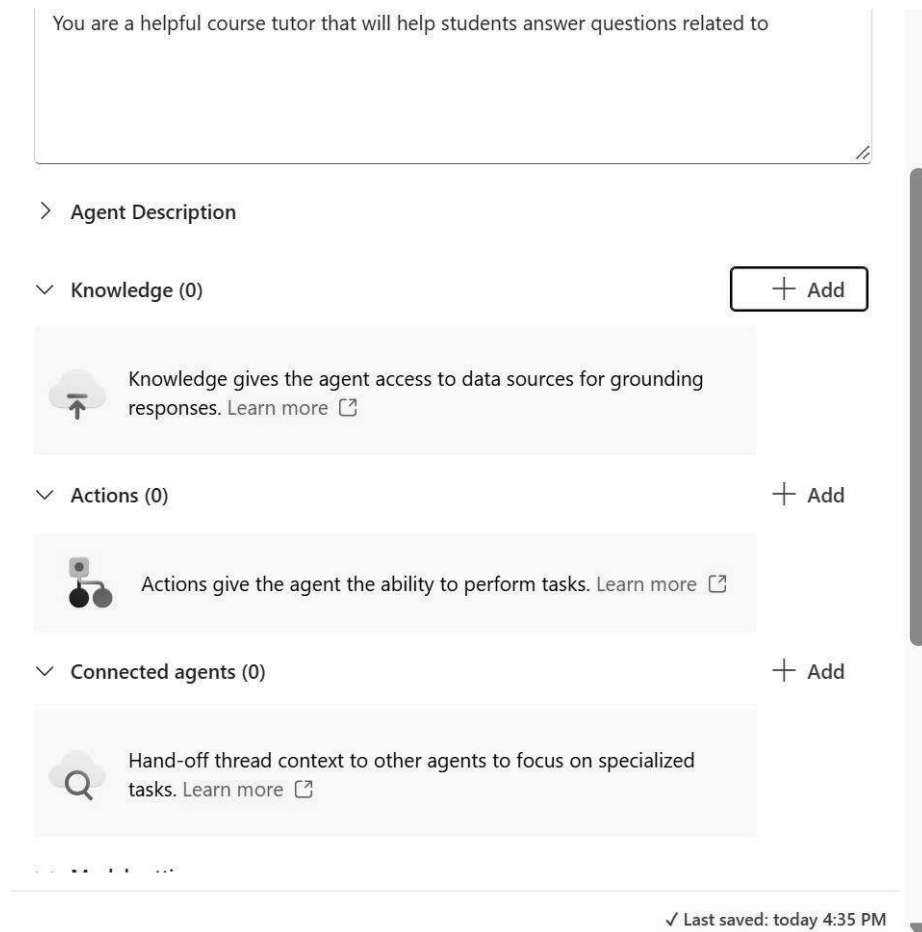
31. We have to select and deploy a model for our agent. Select a model from the available model deployments. Select “gpt-4o-mini” that you created earlier and click “Next”



32. You will now be navigated to the “My agents” screen



33. Select your agent & in the right navigation, go to Knowledge & click on “Add”



34. Select “files” to add your PDF document. Upload your file and click on “Upload and save”. Your PDF will then be added to the vector store.

Adding files

← Back to select knowledge type

Add your own files to augment the agent with specialized knowledge. Files will be added to a vector store that the search index can reference for quick information retrieval. Encrypted or sensitivity-labeled files are not supported. [Learn more](#)

Vector store *

Create a new vector store

Add files *

Upload local

Name

AgentVectorStore_13451

Select local files

File types supported: .c, .cs, .cpp, .doc, .docx, .html, .java, .json, .md, .pdf, .php, .pptx, .py, .rb, .tex, .txt, .css, .js, .sh, .ts

Name		Status	Error	Size	File type	Uploaded ↓
Agentic RAG MavenCourse.pdf		⌛ Not Started		722.41 KB	Local	Jun 2, 2025 4:37 ...

< Prev

Next >

Upload and save

Cancel

35. Now it is time to test your model in the playground. Hit the “Try in Playground button”

Agents

My agents

My threads

Catalog

PREVIEW

+ New agent

Edit connected resources

Refresh

Delete

Copy

Search

Filter

Name	ID	Model	Created	Descr
AgenticRAGCourseHelper	asst_fpRvMSd0UpCI9PtnL4ZioHJ	gpt-4o	Jun 1, 2025 9...	
TariffPromptHelper	asst_ToSI8f8uzmLRyfmHAdWBka9i	gpt-4o	May 27, 2025 ...	

Setup

Hide

Try in playground

Agent id

asst_fpRvMSd0UpCI9PtnL4ZioHJ

Agent name

AgenticRAGCourseHelper

Azure OpenAI resource connection

aicohub6257433964_aoadi

Deployment

gpt-4o (version:2024-11-20)

Create new deployment

Instructions

You are a helpful agent that will help my users on answering questions related to AGENTIC RAG.

Agent Description

Knowledge (1)

Add

36. Write a query in the user query box and see how your agent responds to your question based on the knowledge base. We typed in to tell us about Agentic RAG as that was our document that we uploaded.

Agents playground

+ New agent

</> View code

Delete

Edit connected resources

Thread: thread_uuycdfNZ9iHkoPYvsw2TTDAE 7033t

Can you tell me what is Agentic RAG?

Agentic Retrieval-Augmented Generation (Agentic RAG) is an advanced AI paradigm where large language models (LLMs) autonomously plan and execute iterative processes while retrieving information from external sources. Unlike traditional Retrieval-Augmented Generation (RAG), which follows a static retrieval-read pattern, Agentic RAG involves a dynamic cycle of LLM calls, tool or function usage, and evaluation of results. This iterative, "maker-checker" style improves the accuracy and quality of responses by refining queries and integrating multiple tools like vector searches, SQL databases, or APIs until a satisfactory solution is reached.

The defining characteristic of Agentic RAG is its autonomy in decision-making across these steps, allowing it to own its reasoning process. Traditional RAG systems typically depend on pre-defined paths scripted by humans, whereas Agentic RAG independently determines the sequence of actions based on the information quality. This enables the system to handle complex, multi-step tasks with greater adaptability and efficiency^{1 2}.

If you'd like to explore practical use cases or specific features, let me know!

1 Agentic RAG MavenCourse.pdf

2 Agentic RAG MavenCourse.pdf

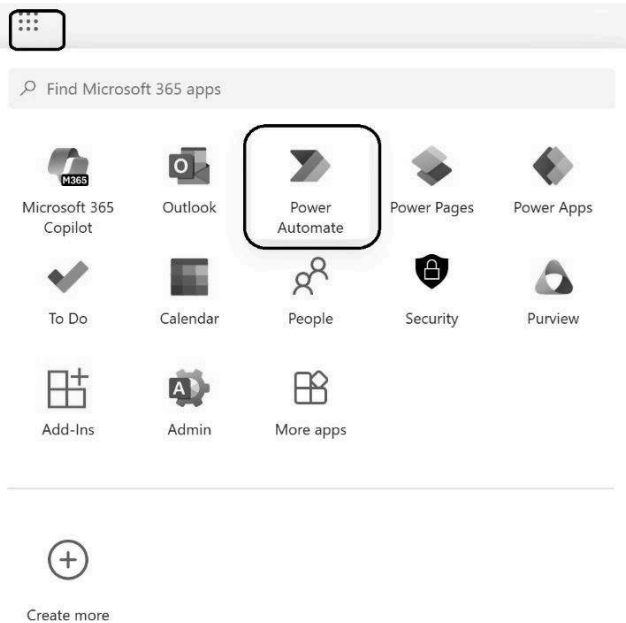
Type user query here. (Shift + Enter for new line)

Messages in the Agents playground are visible to anyone with access to this resource and using the API.

Powerful Agentic Workflows Using Power Automate

Note: This should only be developed after you have successfully created your Agent in Azure AI Foundry with relevant information and role you want the Agent to assume.

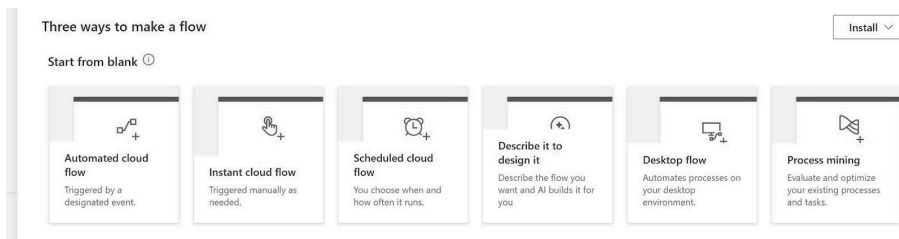
1. Go to your Microsoft365 Account:
 - a. <https://m365.cloud.microsoft/>
2. Once you are logged in, go to select Apps and click on PowerAutomate



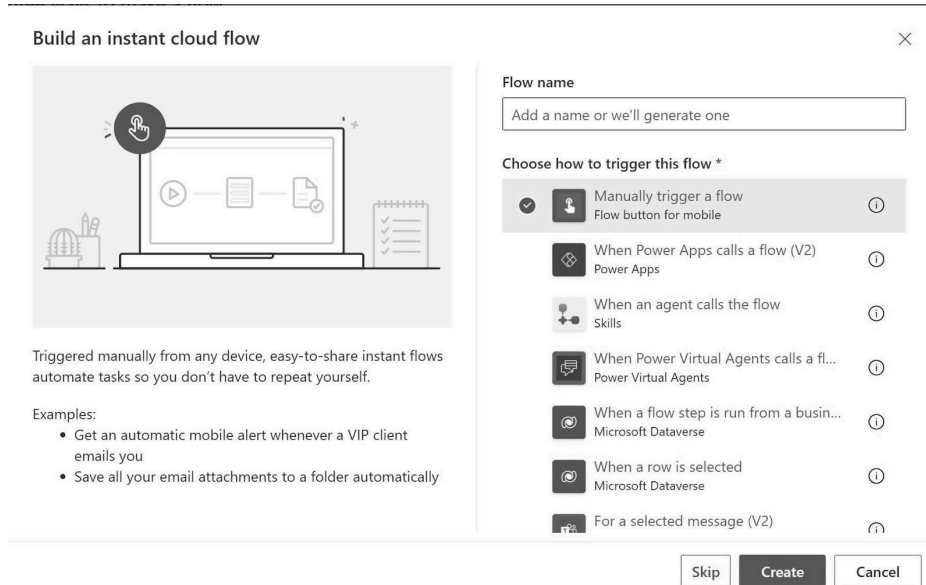
3. Now select Create as we will be creating a New Flow.



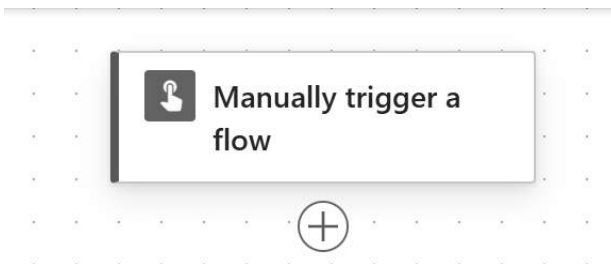
4. Now select Instant Cloud Flows as we for today we will be building one that we want to trigger manually.



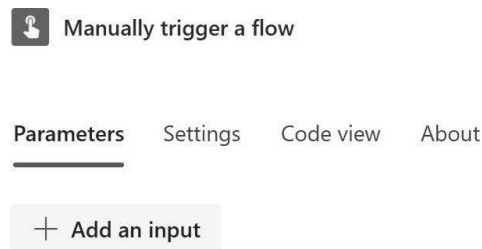
5. We'll select a Manual Trigger for simplicity and click on create



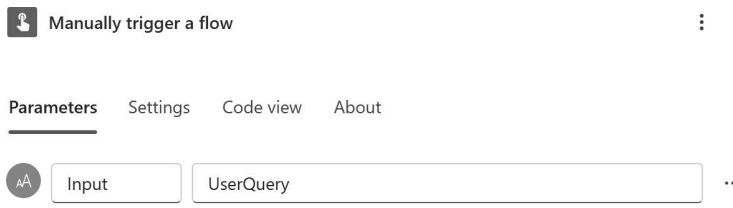
6. Click on Manually Trigger a flow button box



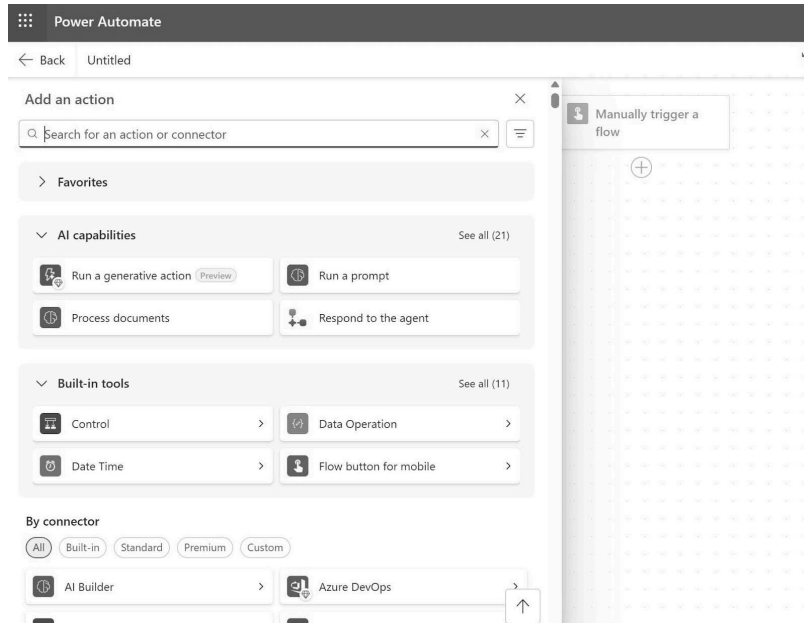
7. Select on Add and Input and select Text



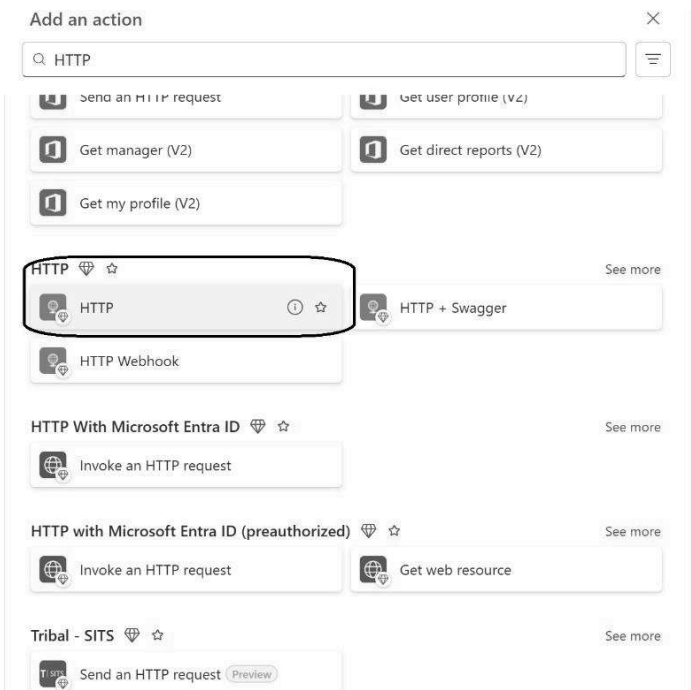
8. Type in the words "UserQuery"



9. Click outside so it saves automatically. Then click on the + button under the Manually trigger a flow and we will add an action, search for HTTP



10. Select the HTTP option where you see a diamond



11. Now let's configure the HTTP option on how it will establish the connection with the agent you built

" to insert dynamic value or expression'. On the right, there is a visual flow diagram with a 'Manually trigger flow' step and an 'HTTP' step, both with plus icons for adding more steps."/>

HTTP

Parameters Settings Code view About

URI *

Enter request url

Press "/" to insert dynamic value or expression

Headers

Enter key Enter value

Queries

Enter key Enter value

Body

Enter request content

Cookie

Enter HTTP cookie

Advanced parameters

Showing 0 of 1 Show all Clear all

Manually trigger flow

HTTP

12. In the URL, enter the endpoint URL from your Agent.

- a. In order to get there, in Azure Foundry AI, go to Models and Endpoints as seen in the screen below:

Manage deployments of your models and services

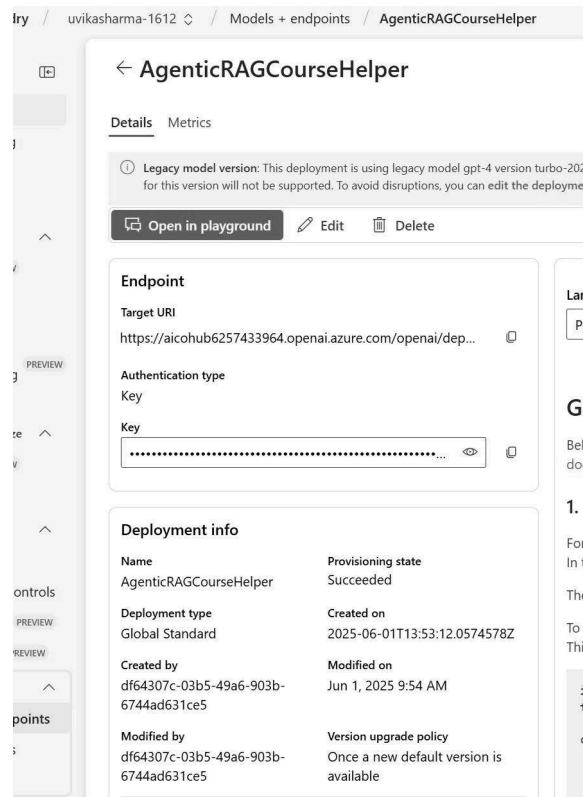
Model deployments Service endpoints

+ Deploy model Refresh Edit Open in playground Reset view

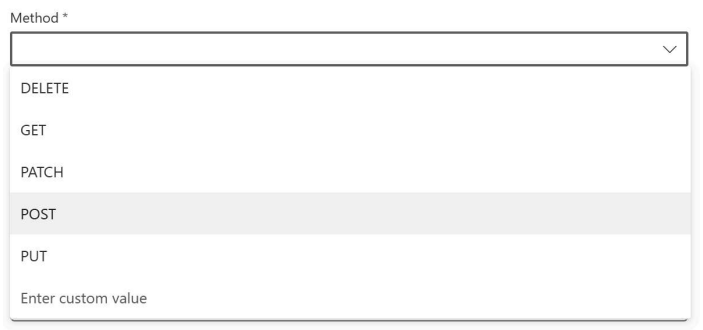
Name	Model name	Model version	State	Model retirement date	Content filter
alcohub6257433964_aioai Azure AI Services	Get endpoint				
AgenticRAGCourseHelper	gpt-4	turbo-2024-04-09	Succeeded		DefaultV2
gpt-4o	gpt-4o	2024-11-20	Succeeded		DefaultV2

Overview Model catalog Playgrounds AI Services Build and customize Agents Templates Fine-tuning Content Understanding Prompt flow Observe and optimize Tracing Monitoring Protect and govern Evaluation Guardrails + controls Risks + alerts Governance My assets Models + endpoints Data + indexes Web apps

- b. Select gpt-4o and then copy the URL address listed there and paste it in the workflow URL value field



13. Select the Method as Post



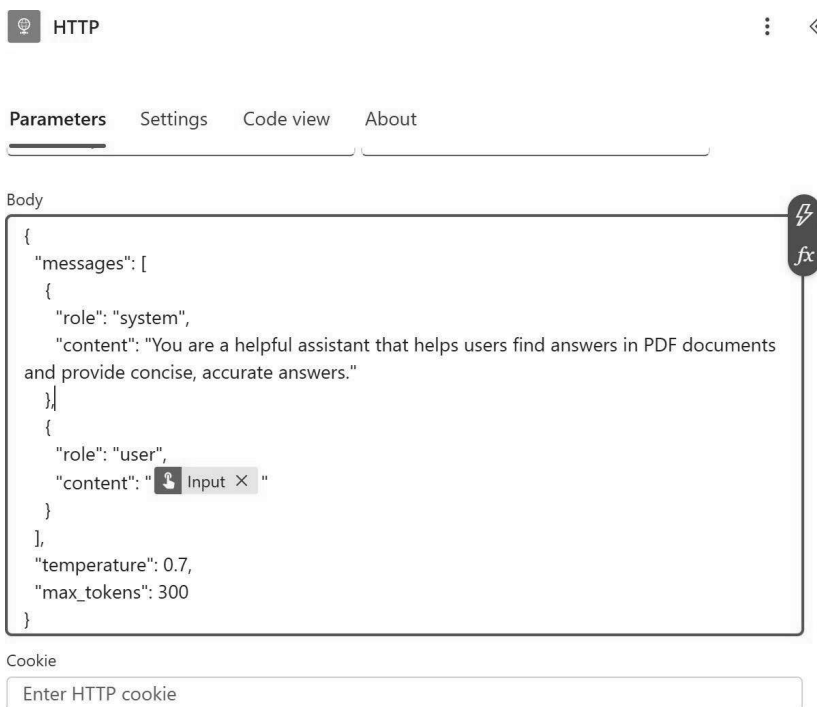
14. In Headers we will populate two values:

- a. Content-Type will be the header name and for value you will put application/json

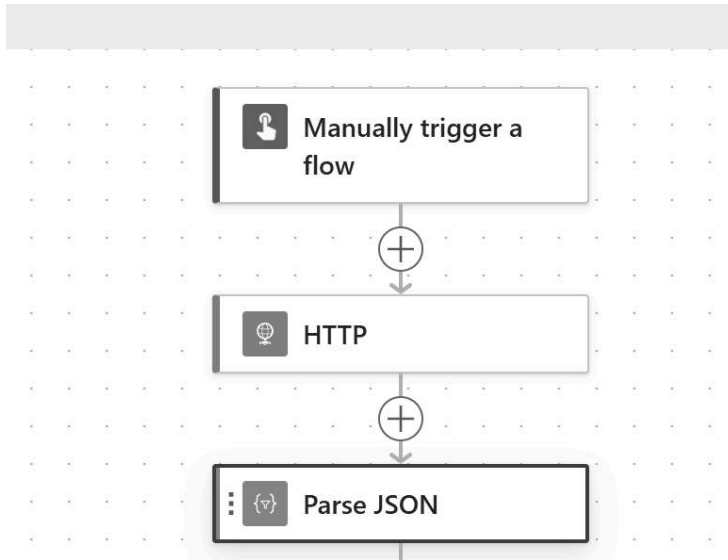

```

{
  "messages": [
    {
      "role": "system",
      "content": "You are a helpful assistant that helps users find answers in PDF documents and provide concise, accurate answers."
    },
    {
      "role": "user",
      "content": "Input"
    }
  ],
  "temperature": 0.7,
  "max_tokens": 300
}

```

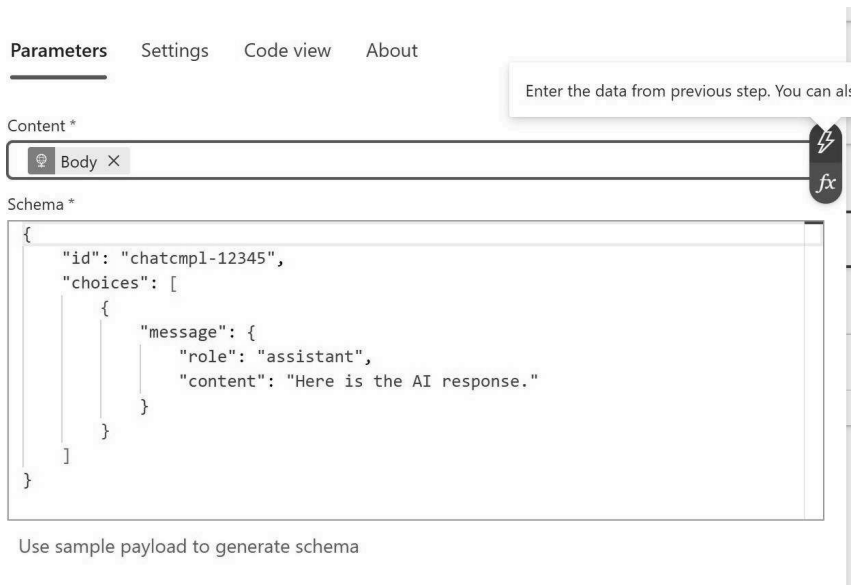


16. Once your HTTP is configured. Now we will Go back to our flow and hit the + button under HTTP and select JSON

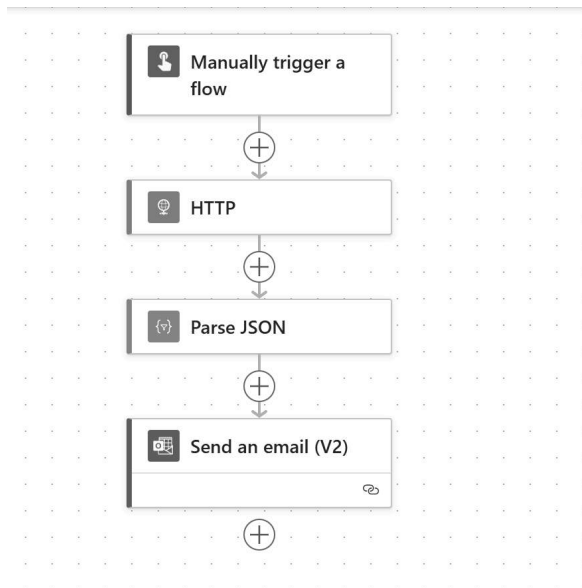


17. Let's configure the JSON Node now. In the Content, look for the Body tag under HTTP using the lightening button shown and in the Schema, value, enter the following text:

```
{
  "id": "chatcmpl-12345",
  "choices": [
    {
      "message": {
        "role": "assistant",
        "content": "Here is the AI response."
      }
    }
  ]
}
```

18. Now let's add the last node by clicking the + icon under JSON and select Send an email (V2)



19. Let's configure the Send an email node now. In the To field, specify where you want to send an email and also specify the subject, and then for body, we will select the fx function button, where we are telling the powerflow to take the parsed json output from the agent and plug it inside the body.

Send an email (V2)

Parameters Settings Code view Testing About

To *

uvika.shu...
Enter part of a name or email address to find more people

Subject *

Your response from Powerautomate

Body *

Normal Arial 15px B I U A G E

Specify the body of the mail

Press '/' to insert dynamic value or expression

Advanced parameters
Showing 1 of 7 Show all Clear all

20. Enter the following text as is in the place shown in text and then hit add.

```
outputs('Parse_JSON')?['body']?['choices'][0]['message']['content']
```

Send an email (V2)

Parameters Settings Code view Testing About

To *

uvika.shu...
Enter part of a name or email address to find more people

Subject *

Your response from Powerautomate

Body *

Normal Arial 15px B I U A G E

Specify the body of the mail

Advanced parameters
Showing 1 of 7 Show all Clear all

Importance
Normal

Function Dynamic content

Search

String functions

concat(text_1, text_2, ...)
Combines any number of strings together

substring(text, startIndex, length?)
Returns a subset of characters from a string.

slice(text, startIndex, endIndex?)
Returns a section of a string defined by the start index and the end index

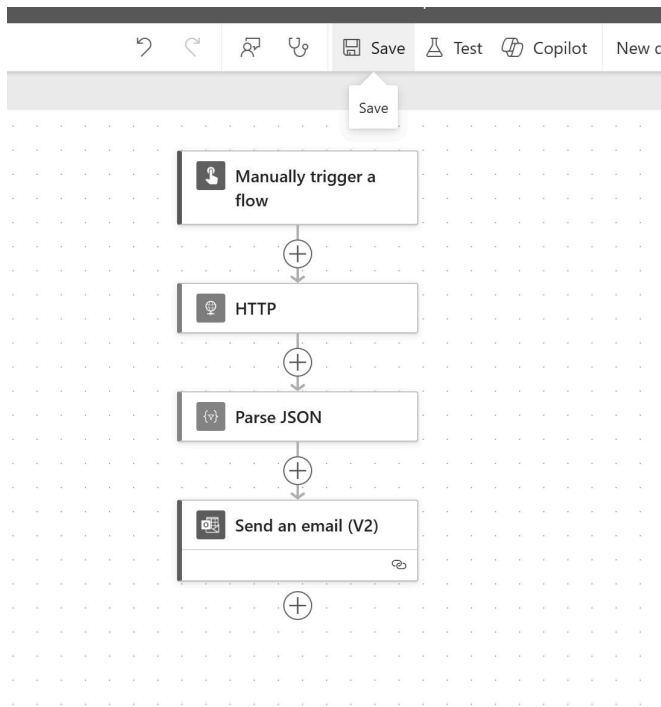
replace(text, oldText, newText)

Add

21. Final Body should look like this



22. Save your Flow



23. Now Let's test our flow, next to save button, click on Test and select Test Flow Manually

Test Flow

- ☒ Manually
Perform the starting action to trigger it.
- ☐ Automatically
There are no runs for this flow.

24. And you will get an info that looks like this as I'm granting permission to use my outlook

Run flow



Button -> HTTP,Parse JSON,Send an email (V2)

Owner: Uvika Sharma

Sign in *

This flow uses the following apps. A green check means you're ready to go.



Office 365 Outlook

Permissions



Continue

Cancel

25. Here you write a query that you want the agent to look for in your document that you had uploaded into your agent in AzureAI Foundry

Run flow



Button -> HTTP,Parse JSON,Send an email (V2)

Owner: Uvika Sharma

Input *

Input is required

This flow uses Office 365 Outlook.

[Review connections and actions](#)

Run flow

Cancel

26. You should get a final message that looks like this and if all worked then you should have received an email. Or else we will go to the Flow Runs page to troubleshoot any errors we received:

Run flow

X

0

Your flow run successfully started. To monitor it, go to the [Flow Runs Pag.](#)

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